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Efficacy of Cryoneurolysis on Chronic Pain in Patients with Knee Osteoarthritis: A Double-blinded Randomized Controlled Sham Trial

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ABSTRACT

Background: The objective was to investigate whether cryoneurolysis was superior to sham in reducing pain intensity in patients with chronic knee osteoarthritis. The hypothesis was that cryoneurolysis is an effective and safe therapy to reduce chronic pain in patients with knee osteoarthritis.

Methods: The study was randomized, double blind, and sham controlled. A total of 87 patients with chronic knee osteoarthritis were randomized to either cryoneurolysis or a sham procedure, followed by an exercise program. The target nerves, anterior femoral cutaneous nerve and infrapatellar branch of the saphenous nerve were identified using transcutaneous electrical nerve stimulation, and cryoneurolysis was performed with ultrasound guidance. The primary outcome was the difference in average pain at 14 days after intervention between the cryoneurolysis and sham group.

Results: For the intention-to-treat analysis, no difference was observed for the primary outcome (0.49; 95% CI, –0.3 to 1.2; $P = 0.198$). Both groups showed a significant reduction in pain scores 14 days after intervention (cryoneurolysis: –1.9; 95% CI, –2.4 to –1.3; and sham: –1.4; 95% CI, –1.9 to –0.8). At 6-month follow-up, a significant reduction in pain was observed in the cryoneurolysis group compared to the sham group (1.1; 95% CI, 0.3 to 1.9; $P = 0.009$). Cryoneurolysis was well tolerated, with minor, transient adverse effects.

Conclusions: This study did not show a clinically relevant reduction in pain after intervention comparing cryoneurolysis and sham. Further studies are needed.

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EDITOR'S PERSPECTIVE

What We Already Know about This Topic

- Cryoablation is a technique that targets the peripheral nerves to provide pain relief
- The efficacy of cryoablation for managing chronic osteoarthritis pain remains unknown

What This Article Tells Us That Is New

- This randomized, double-blinded, sham-controlled study in 87 adults with painful knee osteoarthritis compared cryoablation of the anterior femoral cutaneous and infrapatellar branch of the saphenous nerves directed *via* nerve stimulation and ultrasound *versus* sham ablation, with both groups receiving a postintervention exercise program
- No significant differences in pain severity, functional measures, or related outcomes were noted between groups at 2 weeks after treatment. Although minor 6-month improvements were observed, they were not robust or consistent over time

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Knee osteoarthritis is a prevalent musculoskeletal disorder characterized by degenerative changes, chronic pain, stiffness, and functional disability, affecting millions of patients globally.[1,2] Knee osteoarthritis often leads to severe joint dysfunction, significantly reducing quality of life and resulting in a notable socioeconomic burden due to increased healthcare costs and reduced work productivity.[3,4]

Traditional therapeutic strategies for knee osteoarthritis focus on pain management and the restoration of joint function, offering some symptomatic relief.[5] These strategies include oral analgesics, intra-articular injections, physical therapy, and, in advanced stages, surgical interventions such as arthroplasty. However, the observed benefits often remain moderate, with patients experiencing persistent pain.[6,7] Moreover, surgical approaches, while beneficial for end-stage disease, involve risks and prolonged recovery periods, and 15 to 20% of patients experience persistent pain postoperatively.[2,8]

Given these challenges, innovative treatments are needed. Cryoneurolysis, a minimally invasive procedure, has been posited as a potential treatment for chronic pain conditions.[9,10] Cryoneurolysis involves applying freezing temperatures, ranging from −20° to −100°C, directly to a target peripheral nerve. This process leads to Wallerian degeneration, effectively disrupting the nerve's function. Research has demonstrated the potential benefits of cryoneurolysis in clinical settings such as lumbar facet joint pain,[11] plantar fasciitis,[12] and postthoracotomy pain syndrome[13] with few adverse effects.

For knee osteoarthritis, the role of cryoneurolysis remains to be fully elucidated. While preliminary data suggest its potential in attenuating pain and improving joint function,[14,15] rigorous evidence from randomized controlled trials is essential. In addition, the optimal freezing protocol for cryoneurolysis remains to be outlined. This involves the number of cycles, duration, intensity, and thaw time, which also depends on crucial factors such as nerve target identification, probe placement, and anatomical location of the target tissue. Furthermore, there is a lack of comprehensive research exploring the impact of cryoneurolysis in combination with a structured education and exercise initiative like Godt Liv med Artrose I Danmark (Good Life with osteoArthritis in Denmark; GLA:D) for knee osteoarthritis patients.

In this study, we introduce a unique approach, using transcutaneous electrical nerve stimulation and anatomic landmarks for nerve identification and ultrasound-guided cryoneurolysis. The method considers the inherent variation in anatomical structures between patients and allows for improved patient safety and treatment efficacy. In addition, the current study combines cryoneurolysis treatment with an exercise program known as GLA:D. GLA:D is a standardized education and neuromuscular exercise program designed for managing knee and hip osteoarthritis. Introducing cryoneurolysis as a pain-modulation technique before GLA:D could offer substantial pain reduction. This, in turn, might lead to better program adherence, exercise efficacy, and sustained benefits.

This study's primary aim is to investigate whether cryoneurolysis is superior to sham in providing an initial reduction in pain in patients with chronic knee osteoarthritis 14 days after intervention. Secondarily, the study aims to explore the safety and efficacy of cryoneurolysis or sham in combination with the GLA:D on pain and function after 6 and 12 months. We hypothesized that cryoneurolysis is an effective therapy to reduce pain in patients with chronic knee osteoarthritis.

Materials and Methods

Study Design and Setting

A parallel-group randomized controlled trial was carried out at the Department of Neurology or the University Hospital of Southern Denmark (Esbjerg, Denmark) from July 2019 to March 2023. The principal investigator was Carsten Kock-Jensen. All activities were conducted in alignment with the Declaration of Helsinki and received approval from the Regional Committees on Health Research Ethics for Southern Denmark (approval No. S-20180089). The study was also listed in ClinicalTrials.gov under NCT03774121 on December 3, 2018. The study protocol adhered to the recommendations for Interventional Trials (SPIRIT) and used the Consolidated Standards of Reporting Trials of Nonpharmacological Treatments (CONSORT NPT). Language and grammar in the article were refined on occasion using ChatGPT, an artificial intelligence-based writing aid. The study protocol and further methodologic details can be reviewed in a previous article.[16] The protocol was not accurately described on clinicaltrials.gov at the initiation of the study. This was corrected upon realization in December 2020. This included the following: (1) For practical reasons, the primary outcome was changed from the visual analog scale for pain to the numeric rating scale for pain. (2) The timepoint for the primary outcome was changed from "immediately postintervention" to "14 days postintervention." It was not possible to evaluate pain immediately postintervention because the procedure was performed using local anesthesia, affecting pain perception for at least 12 h. (3) Finally, the evaluation after GLA:D was changed from "3 months" to "after GLA:D" to allow for a sufficient period of time for the patients to complete the program. Note that all procedures were performed according to the study protocol, as described herein, for all patients throughout.

The clinical data were stored on OPEN, the open patient data explorative network of Odense University Hospital (Region of Southern Denmark, with servers in the Region of Southern Denmark through the Research Electronic Data Capture [REDCap] system), and patient-reported outcomes were stored and captured using the PainData registry (Pain Center, Odense University Hospital; Department of Anesthesiology and Intensive Care Medicine, Odense

University Hospital). This system ensured encrypted connections with restricted access and met data security requirements. The project was reported to the Danish Data Protection Agency (https://www.datatilsynet.dk/english) and was managed in line with the Data Protection Act 2018, c. 12. (available at: https://www.legislation.gov.uk/ukpga/2018/12/contents/enacted). Only lead investigators had access to the complete data set.

Participant Enrollment and Recruitment Process

Patients with knee osteoarthritis pain referred to GLA:D by their primary care physician, before surgical eligibility assessment at the hospital, were considered for this study. Patients were made aware of the study through public posts and at the offices of primary care physicians, physiotherapists, and others. Eligible candidates who consented to participate were screened by the primary investigator against the specified eligibility criteria, and an orthopedic surgeon confirmed the diagnosis of knee osteoarthritis and related pain. Eligible patients moved onto a secondary visit, during which the nerves around the knee were identified using ultrasound and electrical nerve stimulation, followed by a diagnostic genicular nerve block. Patients who reported a reduction in knee pain intensity of 50% or more due to the nerve block were then randomized into the cryoneurolysis treatment group (cryo) or the sham intervention group (sham).

Eligibility Criteria. Participants were eligible for inclusion if they were referred to the GLA:D program,[17] were aged 18 yr or older, and had experienced knee pain for more than 6 months with a pain intensity of at least 4 on the numeric rating scale. All participants required radiographic evidence of osteoarthritis corresponding to Kellgren–Lawrence grades 2 to 4,[18] and a diagnostic genicular nerve block needed to produce at least a 50% reduction in their knee pain. Additionally, participants had to demonstrate both written and oral comprehension of Danish.

Individuals were excluded if they had a history of systemic inflammatory conditions or had previously undergone cryoneurolysis for the knee. Those who had received hyaluronic acid injections within the last 30 days or corticosteroid injections within 3 months were not eligible. Other reasons for exclusion included structural abnormalities unrelated to osteoarthritis that affected locomotion or knee function, a body mass index less than 18 or above 40 kg/m^2, ongoing treatment for other pain conditions, pregnancy, coagulopathy, uncontrolled serious illnesses (*e.g.*, cancer or diabetes), and diseases associated with adverse reactions to cold.

Randomization Process

Participants were randomized using a computer-generated block randomization method with a 1:1 allocation ratio, incorporating random block sizes of 2, 4, and 6. An external coinvestigator handled the sequence creation while ensuring allocation concealment. Blinding was implemented for patients, therapists, and data managers. The surgeon and nurse performing the procedure were not blinded. If deemed necessary by investigators and physicians, unblinding was executed following established emergency procedures, ensuring the study's integrity and confidentiality. Emergency unblinding did not occur.

Interventions

Cryoneurolysis and Sham. For the first visit, radiologic confirmation and clinical assessment of knee osteoarthritis as the pain source was followed by identifying the infrapatellar branch of the saphenous nerve and anterior femoral cutaneous nerve. Predefined areas, based on previous studies,[18,19] were identified for each nerve. Areas where the patient indicated a response to a transcutaneous electrical nerve stimulation probe were marked and subsequently rechecked with progressively lower current to accurately pinpoint the nerve location. The location was visually confirmed using ultrasound. Nerve structures appear as hypoechoic nerve fascicles within the hyperechoic epineurium, creating a honeycomb-like pattern in a short-axis view. An ultrasound-guided diagnostic nerve block using ropivacaine (5 mg/ml), 1 to 2 ml, was administered perineurally adjacent to each nerve. Patients reporting a 50% or greater reduction in knee pain score, using the numeric rating scale for pain, were scheduled for a second visit for cryoneurolysis.

For the second visit, before treatment, ropivacaine (5 mg/ml) was injected locally at the insertion point, 4 to 6 cm from target nerve locations, allowing for continuous patient feedback. The cryoneurolysis probe (Iceseed 1.5; Galil Medical Ltd., USA) was inserted near the target nerve, with ultrasound guiding to accurately determine nerve location, considering adjacent neurovascular structures and anatomic structure variations. Cryoneurolysis involved a single freeze cycle for 30 s at 20% effect, followed by 2.5 min at 60% effect. To avoid frostbite, heating pads were placed on top of the skin to counteract the freezing temperatures from below, and hydrodissection (saline injection to separate tissue) was used to increase the distance between cutaneous tissue and the freezing temperatures. After the cycle, 1 min of active thaw and 1 min of passive thaw were used. VisualICE (Galil Medical Ltd.) was the machine used for the intervention, utilizing argon for cooling and helium for thawing, facilitating reversible nerve destruction, known as Wallerian degeneration.[20–22]

The sham intervention mirrored the cryo procedures but instead of generating freezing temperatures at the target nerve, gas was transferred to a secondary probe inserted in a hidden basin. This procedure ensured an identical procedural experience, including sound, local anesthesia, and similar visible postprocedure marks for both groups.

GLA:D. After the cryoneurolysis or sham intervention, participants from both groups participated in the GLA:D program for a total of 8 weeks, facilitated by specialized physiotherapists at GLA:D-approved clinics. The initial educational phase, three sessions over 2 weeks, focused on osteoarthritis and the benefits of exercise as treatment. After the educational phase, patients continued a 6-week group-based NEuroMuscular Exercise program (NEMEX), comprising 12 sessions, each lasting 60 min. Patients who were unable or chose not to participate in the supervised sessions had the option to perform the program at home.

Outcomes Measurement

Participants underwent evaluations at baseline, at 14 days after study intervention, after completion of GLA:D, and at 6 and 12 months postcryoneurolysis. This included the primary outcome and all secondary outcomes. A planned follow-up at 24 months was cancelled due to project delays. Outcomes between the cryo and sham groups were compared at each designated timepoint.

Primary Outcome. The primary outcome was the difference in average pain at 14 days postintervention between the cryoneurolysis and sham groups. Pain was assessed as the average pain during the last 24 h on an 11-point numeric rating scale, where 0 was "no pain," and 10 was "worst pain imaginable." The numeric rating scale is validated and extensively used in both clinical and research contexts.[23]

Secondary Outcomes. Knee pain was also assessed on the numeric rating scale as the worst and least pain during the past 7 days, as well as during standardized activity (sit-to-stand) and during rest. Functional performance was assessed *via* the 30-s chair-stand test, the 40-m fast-paced walk test, and isometric knee maximal voluntary contraction force (described elsewhere[16]). The planned nine-step stair climb test was dropped due to a change in location during the study period. The 30-s chair-stand test involved repeated sit-to-stand movements for 30 s. Starting seated with feet flat, shoulder-width apart, and arms crossed on the chest, the position changes to standing with fully extended hips and knees, then back to sitting with the bottom fully touching the seat. The outcome was the total number of complete chair stands performed in 30 s (one chair stand was a stand followed by a sit).

The 40-m fast-paced walk test involved walking as fast as possible along a 10-m walkway, turning around a cone/tape, and repeating for a total of 40 m. Usual footwear and regular walking aids are allowed and recorded. The outcome was the time taken (seconds), with timing paused during turns.

Maximal voluntary contraction was assessed by measuring isometric force of the knee extensors. Patients was seated with knees and hips at 90° and with the pelvis and chest restrained by straps. A chain attached to the chair and a force transducer was placed just above the malleolus. Patients performed three maximum knee extensions against the chain, with 1 min rest between each. The highest peak value was recorded as the maximal voluntary contraction force.

Additionally, patient-reported outcomes data were collected using an electronic questionnaire platform (the PainData system[24]), featuring multiple integrated questionnaires such as Knee Injury and Osteoarthritis Outcome Score (KOOS), EQ-5D, Pain Catastrophizing Scale, Patient Health Questionnaire (PHQ9), and Generalized Anxiety Disorder (GAD7). Further details can be reviewed in Nakano *et al.*[16] Adverse effects, defined as any undesirable experience prompting healthcare system contact, were recorded and assessed during follow-ups *via* prespecified symptom inventories and open-ended queries.[16]

Statistical Analysis

Sample Size. Sample size determination was based on the null hypothesis, positing no difference in average pain between the cryoneurolysis and the sham group after intervention. Estimations for a two-sample means test, factoring in a significance level of 0.05 and a common SD of 3 in numeric rating scale pain intensity scores, indicated that a total of 74 individuals were required for the study. This would achieve a power of at least 80% to recognize a minimal clinically important difference of 2 in numeric rating scale pain scores between groups. With an anticipated dropout rate of 20%, the study aimed to include 94 participants, divided equally between groups. The minimal clinically important difference and common SD is based on previous findings with a similar patient group and intervention.[17,25]

Analysis. To evaluate the distributions of the continuous outcomes, visual inspection of the studentized residuals was applied to evaluate whether the assumption of normality was reasonable. An intention-to-treat analysis and a per-protocol analysis were used for all allocated patients. Mixed linear regression with the assumption of unstructured covariance was used to model the effect of the cryoneurolysis treatment over time and to take into account the repeated measures by including individuals as random variables. Time, group of treatment, baseline pain levels, and the group # time interaction was used as fixed effects to estimate change over time. The results are depicted using predicted average values and predicted differences between groups (raw values included in the supplemental digital content, https://links.lww.com/ALN/D845). These values represent the estimated outcomes for each group, accounting for the random effects and covariates included in the model, and are illustrated using marginal effects using the margins command in Stata16. A P value < 0.05 was considered statistically significant. The P value was adjusted for multiple comparisons ($k = 9$) for the mixed effects model

evaluating the primary outcome only, average pain over 24 h on the numeric rating scale, using a Holm–Bonferroni test.[26] Secondary outcomes were not adjusted, because these were explorative.[27]

Results

This study received inquiries from 228 patients. Continued interest in study participation and eligibility were confirmed, after which 122 were invited for screening. A diagnostic nerve block was performed on 87 patients, and all experienced a greater than 50% reduction in pain. A total of 87 patients were thus randomized to either the cryo group (n = 44) or the sham group (n = 43; fig. 1). At 14-day follow-up, 84 patients remained in the study, *i.e.*, a 3.5% dropout rate for the primary endpoint analysis, *i.e.*, 100% for cryoneurolysis (44 of 44) and 93% for sham (40 of 43). Two patients withdrew consent and did not wish to receive treatment, and one no longer experienced pain at baseline and was thus not eligible. A total of 58 patients were available for analysis at GLA:D follow-up (32 of 44 for the cryoneurolysis group and 26 of 43 for the sham group). Of those, 13 patients chose to discontinue GLA:D but remained in the study. At 6- and 12-month follow-up, 68 (37 of 44 in the cryoneurolysis group and 31 of 43 in the sham group) and 63 (37 of 44 in the cryoneurolysis group and 26 of 43 in the sham group) remained in the study. Recruitment stopped at 87 patients for ethical and methodologic concerns because the project was delayed due to challenges with recruitment and COVID-19. Demographic data are displayed in table 1.

Primary Outcome

The primary endpoint was the difference in average 24-h numeric rating scale pain scores at 14 days postintervention in patients that received cryoneurolysis compared to sham.

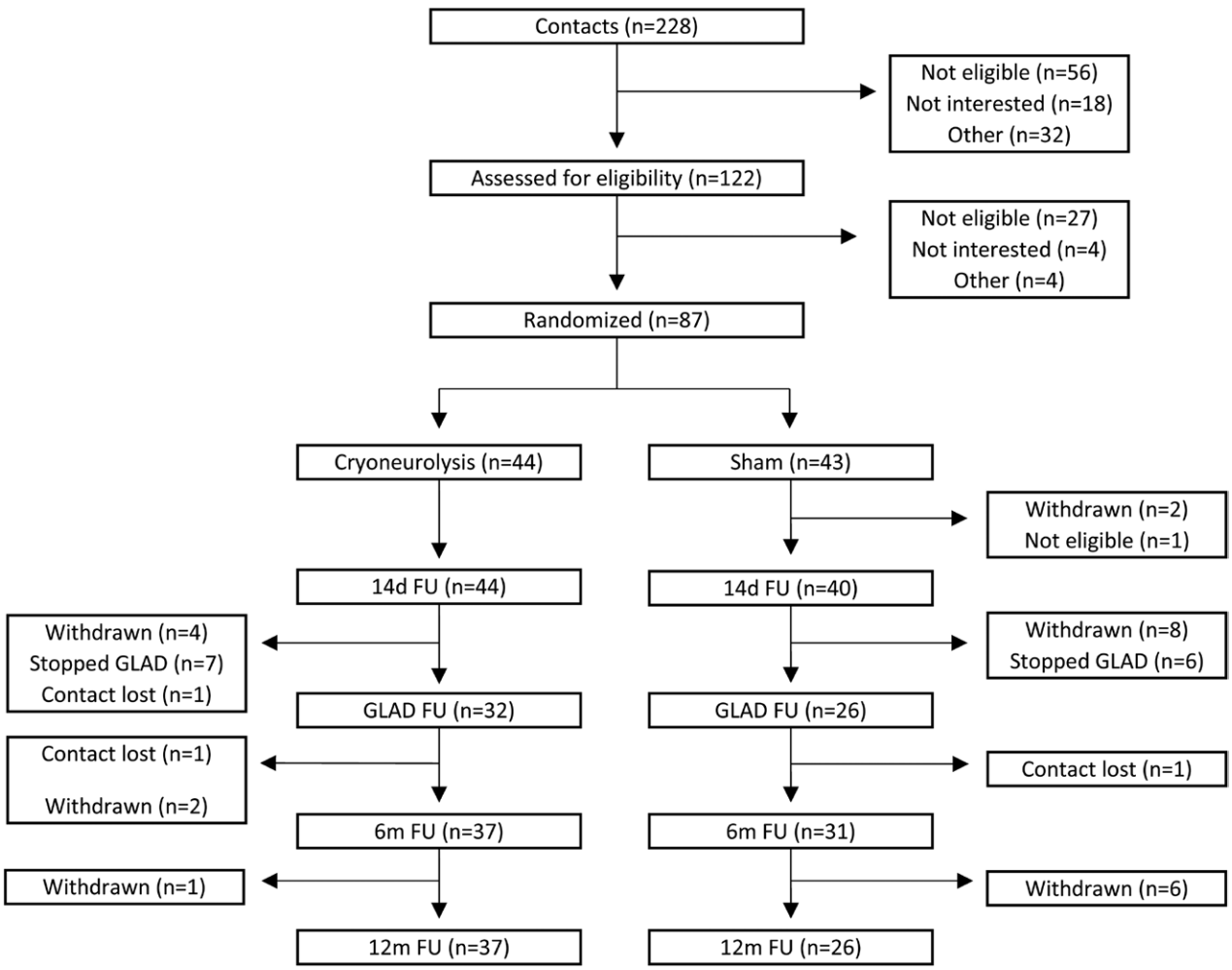

Fig. 1. Study flow chart. Withdrawn, patients chose to stop further participation in the study; stopped GLAD, did not complete the Godt Liv med Artrose i Danmark (Good Life with osteoArthritis in Denmark; GLA:D) program but remained in the study; contact lost, not available but remained in the study. FU, follow-up.

Table 1. Patient Characteristics

Variable	Cryo	Sham
Age, yr		
Mean ± SD	68.6 ± 8.4	62.2 ± 10.7
Sex, n (%)		
Male	24 ± 50.0	24 ± 51.3
Female	20 ± 50.0	19 ± 48.7
Body mass index, kg/m^2		
Mean ± SD	28.7 ± 4.8	29.2 ± 4.9
Baseline NRS, mean ± SD		
24 h	5.3 ± 1.6	4.3 ± 1.4
Active	3.1 ± 2.4	3.2 ± 2.2
Rest	2.1 ± 2.0	2.1 ± 2.0
KOOS, mean ± SD		
Symptoms	59.1 ± 17.3	59.6 ± 15.4
ADL function	62.3 ± 13.4	65.1 ± 14.8
Sport function	27.3 ± 18.2	28.3 ± 18.6
Pain	54.9 ± 11.8	57.6 ± 13.1
QoL	37.8 ± 12.6	39.2 ± 12.1

The average values are displayed with the percentage of the total number of participants in parentheses unless otherwise stated.

ADL, activities of daily living; Cryo, cryoneurolysis; KOOS, Knee Injury and Osteoarthritis Outcome Score; NRS, numeric rating scale; QoL, quality of life.

The results showed a significant reduction in 24-h numeric rating scale pain scores in both groups (cryoneurolysis: −1.9; 95% CI, −2.4 to −1.3; and sham: −1.4; 95% CI, −1.9 to −0.8) and no significant estimated difference between groups (0.49; 95% CI, −0.3 to 1.2; $P = 0.198$; table 2; fig. 2), accounting for the random effects and covariates included in the mixed effects model.

Secondary Outcomes

Numeric Rating Scale. The pain scores continued to show a decline after GLA:D with no difference between groups (0.8; 95% CI, −0.1 to 1.6; $P = 0.064$). After 6 months, the cryoneurolysis group showed a significant reduction in 24-h numeric rating scale pain scores compared to sham (1.1; 95% CI, 0.3 to 1.9; $P = 0.009$), which disappeared at the 12-month follow-up (0.7; 95% CI, −0.2 to 1.5; $P = 0.111$).

Knee pain was also assessed as the worst and least pain during the past 7 days and during activity and rest. Worst pain experienced and pain during rest and activity showed no significant differences between groups at all timepoints. The least pain experienced showed a significant reduction in the cryoneurolysis group at 14 days (1.2; 95% CI, 0.5 to 2.0; $P = 0.002$), after GLA:D (1.5; 95% CI, 0.6 to 2.4; $P = 0.001$), and at the 6-month (1.4; 95% CI, 0.5 to 2.2; $P = 0.002$) follow-up (table 3).

Adverse Effects. No serious adverse effects were reported throughout the study in either group. Several minor adverse effects were observed that were expected and defined *a priori* to study start, including altered sensation, bruising, local pain, numbness, redness, swelling, tenderness on palpation, and tingling. These were observed in both groups albeit significantly more in the cryo group (table 4). Three patients observed a small lesion after treatment corresponding to a superficial frostbite at the level of the skin. None required treatment.

Functional Capacity. There was no significant difference between the cryoneurolysis group and sham group in sit-to-stand or maximal voluntary contraction at either timepoint (table 5). The 40-m walk test did show that the capacity for walking increased in the cryoneurolysis group compared to sham as shown by significantly faster time measurements at 14 days (1.6 s; 95% CI, 0.2 to 3.0; $P = 0.025$), after GLA:D (1.8 s; 95% CI, 0.1 to 3.5; $P = 0.04$), and at 6-month follow-up (2 s; 95% CI, 0.4 to 3.6; $P = 0.015$).

Patient-reported Questionnaire Data. Patient-reported outcomes showed no significant differences for quality of life as indicated by the KOOS scores and EQ-5D scores (tables 6 and 7). The degree to which aspects of life were disrupted (pain-related disability score), depression (Patient Health Questionnaire), pain catastrophizing (Pain Catastrophizing Scale), and fear of movement (Tampa Scale of Kinesiophobia) did not change at either timepoint (table 7). The questions on anxiety, as measured by the generalized anxiety disorder questionnaire, indicated that anxiety was lower in the cryoneurolysis group compared to sham at the 6-month follow-up (1.2; 95% CI, 0.1 to 2.3; $P = 0.039$).

Discussion

This study aimed to investigate the efficacy of cryoneurolysis for the treatment of osteoarthritis-related knee pain. Both the cryoneurolysis and sham group showed a reduction in pain 14 days after treatment with no difference between the two groups. After 6 months, after GLA:D participation, the cryoneurolysis group showed significantly greater pain reduction than sham. These data were supported by a small increased ability to walk and lower least pain scores and anxiety; however, no other changes were observed, which suggests that these differences represent random fluctuations rather than a meaningful treatment effect. When viewed across all assessed timepoints, the results indicate that cryoneurolysis has no meaningful, robust benefit over sham for chronic knee pain. Although a minor improvement emerged at the 6-month follow-up, this difference should be interpreted with caution. Further research is needed.

There were no serious adverse effects in the current study. Several expected mild adverse effects were defined before the study start, among which numbness (49%) and swelling (29%) were observed in the group that received cryoneurolysis and not in the sham group. It should be emphasized that these adverse effects are in line with effects after nerve ablation and could be indicative of the

Table 2. NRS Pain Scores at 24 h

	Predicted Average Values [95% CI]		Predicted Difference over Time [95% CI]			
NRS at 24 h	**Cryo**	**Sham**	**Cryo**	**Sham**	**Time # Group [95% CI]**	***P* Value**
ITT						
Baseline	5.0 [4.5 to 5.5]	4.6 [4.1 to 5.1]				
14 days	**3.2 [2.7 to 3.6]**	**3.2 [2.7 to 3.7]**	**−1.9 [−2.4 to −1.3]**	**−1.4 [−1.9 to −0.8]**	**0.49 [−0.3 to 1.2]**	**0.198**
GLA:D	2.7 [2.2 to 3.3]	3.1 [2.5 to 3.7]	−2.3 [−2.9 to −1.7]	−1.5 [−2.1 to −0.9]	0.8 [−0.1 to 1.6]	0.064
6 months	2.6 [2.1 to 3.1]	3.2 [2.7 to 3.8]	−2.5 [−3.0 to −1.9]	−1.4 [−2.0 to −0.8]	1.1 [0.3 to 1.9]	*0.009*
12 months	3.1 [2.6 to 3.6]	3.4 [2.8 to 3.9]	−1.9 [−2.4 to −1.3]	−1.2 [−1.8 to −0.6]	0.7 [−0.2 to 1.5]	0.111
PP						
Baseline	5.1 [4.5 to 5.6]	4.5 [3.9 to 5.1]				
14 days	2.8 [2.2 to 3.3]	3.3 [2.6 to 3.9]	−2.3 [−2.9 to −1.7]	−1.3 [−1.9 to −0.6]	1.0 [0.2 to 1.9]	*0.022*
GLA:D	2.7 [2.2 to 3.3]	3 [2.4 to 3.6]	−2.3 [−2.9 to −1.7]	−1.5 [−2.2 to −0.8]	0.9 [−0.1 to 1.8]	0.063
6 months	2.6 [2.1 to 3.2]	3.1 [2.5 to 3.7]	−2.5 [−3.1 to −1.8]	−1.4 [−2.1 to −0.7]	1.0 [0.1 to 2.0]	*0.024*
12 months	3.3 [2.7 to 3.8]	3.4 [2.7 to 4.0]	−1.8 [−2.4 to −1.2]	−1.2 [−1.8 to −0.4]	0.7 [−0.2 to 1.6]	0.134

The table shows the predicted average values and differences for the numeric rating scale for pain for the ITT analysis and PP analysis, respectively. These values represent the estimated outcomes for each group, accounting for the random effects and covariates included in the mixed effects model, *i.e.*, baseline pain level, time, group. The interaction "Time # Group" signifies the difference between groups across time. The primary outcome measure is in *bold type*, and statistically significant *P* values were adjusted for the intention-to-treat analysis using the Holm–Bonferroni test ($P < 0.01$). 95% CIs are displayed in brackets, and statistically significant results are *italicized*.

Cryo, cryoneurolysis; GLA:D, Godt Liv med Artrose I Danmark (Good Life with osteoArthritis in Denmark); ITT, intention-to-treat; NRS, numeric rating scale; PP, per-protocol.

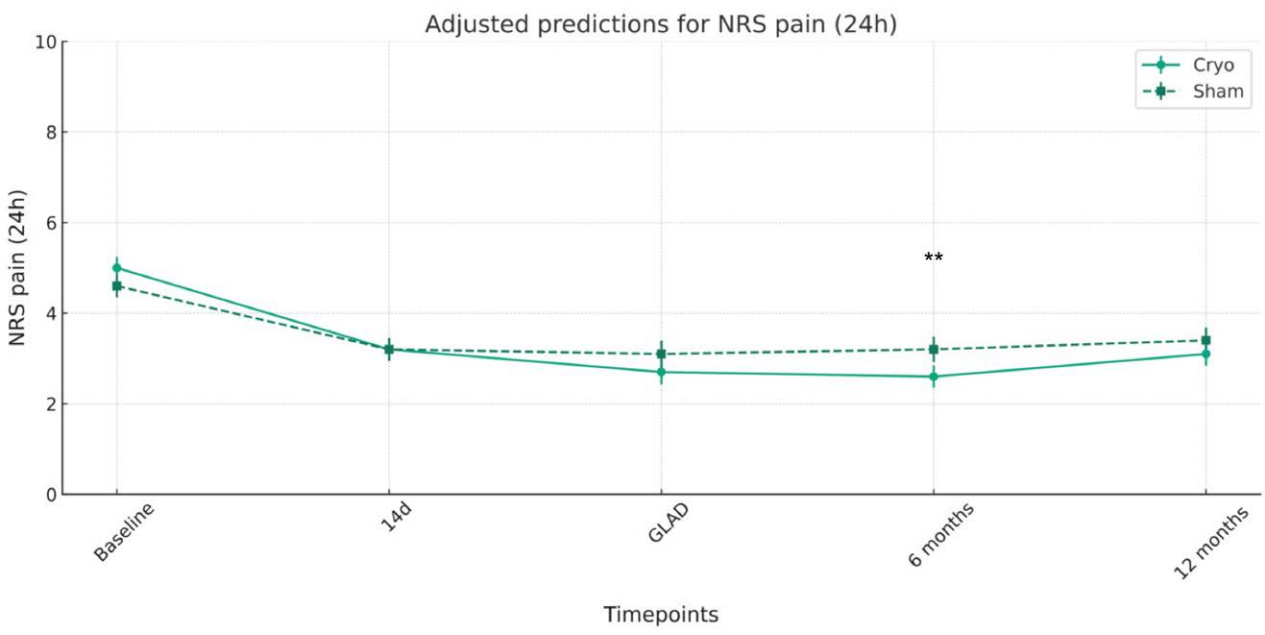

Fig. 2. Predicted average values of an intention to treat analysis for pain rating on the numeric rating scale at each time point for the cryoneurolysis (Cryo; *solid line*) and sham (*dotted line*) groups. $^{**}P = 0.009$; α after Holm–Bonferroni adjustment were $P < 0.01$. The results are depicted using predicted average values and predicted differences between groups. These values represent the estimated outcomes for each group, accounting for the random effects and covariates included in the mixed effects model, *i.e.*, baseline pain level, time, group. GLAD, Godt Liv med Artrose i Danmark (Good Life with osteoArthritis in Denmark); NRS, numeric rating scale.

technical success of the procedure, accurately targeting the nerves. Most adverse effects resolved within 14 days and all resolved by 6 months without requiring treatment. The resolution of numbness might suggest that the protocol caused neuropraxia rather than axonotmesis, leading to reduced analgesia over time. However, the link between numbness and analgesia is unclear, as previous studies showed analgesia and a lower incidence of numbness.[15] The lack of adverse effects is in line with previous studies suggesting reduced risk of damaging structures adjacent to

Table 3. Additional NRS Pain Ratings

	Predicted Average Values [95% CI]		Predicted Difference over Time [95% CI]			
Endpoint	**Cryo**	**Sham**	**Cryo**	**Sham**	**Time # Group [95% CI]**	***P* Value**
NRS active						
Baseline	3.1 [2.6 to 3.5]	3.1 [2.6 to 3.6]				
14 days	1.7 [1.2 to 2.1]	1.9 [1.4 to 2.4]	−1.4 [−2.0 to −0.9]	−1.2 [−1.8 to −0.6]	0.2 [−0.6 to 1.1]	0.588
GLA:D	1.2 [0.6 to 1.8]	1.6 [0.9 to 2.2]	−1.9 [−2.6 to −1.2]	−1.5 [−2.3 to −0.8]	0.3 [−0.7 to 1.3]	0.506
6 months	1.2 [0.7 to 1.8]	1.8 [1.2 to 2.3]	−1.9 [−2.5 to −1.2]	−1.4 [−2.0 to −0.7]	0.5 [−0.5 to 1.4]	0.304
12 months	1.7 [1.2 to 2.2]	1.8 [1.2 to 2.4]	−1.4 [−2.0 to −0.8]	−1.3 [−2.0 to −0.6]	0.1 [−0.9 to 1.0]	0.910
NRS rest						
Baseline	2.1 [1.7 to 2.5]	2.1 [1.7 to 2.5]				
14 days	1.1 [0.7 to 1.5]	0.8 [0.4 to 1.2]	−1.0 [−1.5 to −0.4]	−1.3 [−1.9 to −0.7]	−0.3 [−1.1 to 0.5]	0.434
GLA:D	0.4 [−0.1 to 0.9]	0.8 [0.2 to 1.3]	−1.7 [−2.3 to −1.1]	−1.4 [−2.0 to −0.7]	0.3 [−0.6 to 1.2]	0.528
6 months	0.6 [0.2 to 1.1]	0.7 [0.2 to 1.2]	−1.5 [−2.1 to −0.9]	−1.4 [−2.1 to −0.8]	0.0 [−0.8 to 1.0]	0.932
12 months	1.0 [0.6 to 1.5]	1.1 [0.6 to1.7]	−1.1 [−1.6 to −0.5]	−1.0 [−1.7 to −0.3]	0.1 [−0.8 to 0.9]	0.898
NRS least						
Baseline	2.6 [2.2 to 3.1]	2.0 [1.5 to 2.4]				
14 days	1.2 [0.8 to 1.7]	1.8 [1.3 to 2.3]	−1.4 [−1.6 to −0.8]	−0.1 [−0.7 to 0.4]	1.2 [0.5 to 2.0]	*0.002*
GLA:D	0.7 [0.1 to 1.2]	1.5 [0.9 to 2.1]	−2.0 [−2.6 to −1.3]	−0.5 [−1.2 to 0.2]	1.5 [0.6 to 2.4]	*0.001*
6 months	1.3 [0.8 to 1.8]	2.0 [1.5 to 2.6]	−1.3 [−1.9 to −0.7]	0.1 [−0.6 to 0.7]	1.4 [0.5 to 2.2]	*0.002*
12 months	1.9 [1.4 to 2.5]	1.5 [0.9 to 2.1]	−0.7 [−1.3 to −0.1]	−0.4 [−1.1 to 0.2]	0.2 [−0.7 to 1.1]	0.607
NRS strong						
Baseline	6.2 [5.6 to 6.8]	6.0 [5.4 to 6.6]				
14 days	4.3 [3.7 to 4.9]	4.2 [3.5 to 4.8]	−1.9 [−2.7 to −1.1]	−1.8 [−2.6 to −1.0]	0.1 [−1.1 to 1.2]	0.924
GLA:D	3.6 [2.9 to 4.4]	3.8 [3.0 to 4.6]	−2.6 [−3.5 to −1.7]	−2.2 [−3.2 to −1.3]	0.4 [−0.9 to 1.7]	0.580
6 months	3.4 [2.8 to 4.1]	3.5 [2.8 to 4.3]	−2.8 [−3.6 to −2.0]	−2.5 [−3.4 to −1.6]	0.3 [−0.9 to 1.5]	0.614
12 months	3.9 [3.2 to 4.6]	4.4 [3.6 to 5.2]	−2.4 [−3.2 to 1.5]	−1.6 [−2.6 to −0.7]	0.7 [−0.5: 2.0]	0.265

The table shows the predicted average values and differences for the NRS for pain for the ITT analysis. These values represent the estimated outcomes for each group, accounting for the random effects and covariates included in the mixed effects model, *i.e.*, baseline pain level, time, group. The interaction "Time # Group" signifies the difference between groups across time. "NRS active" indicates the pain measured during activity (sit-to-stand), "NRS rest" shows pain measured during rest, "NRS least" is the least pain experienced for the past week, and "NRS strong" is the strongest pain experienced for the past week. 95% CIs are displayed in brackets, and statistically significant *P* values ($P < 0.05$) are *italicized*.

Cryo, cryoneurolysis; GLA:D, Godt Liv med Artrose I Danmark (Good Life with osteoArthritis in Denmark); ITT, intention-to-treat; NRS, numeric rating scale.

Table 4. Frequency of Observed Adverse Effects

	n (%)		
Adverse Event	**Cryo (n = 3)**	**Sham (n = 44)**	**Total (n = 87)**
Expected			
Altered sensation	4 (9.3)	1 (2.3)	5 (5.8)
Bruising	12 (27.9)	11 (25.0)	23 (26.4)
Hyperpigmentation	0 (0.0)	0 (0.0)	0 (0.0)
Crusting	0 (0.0)	0 (0.0)	0 (0.0)
Itching	0 (0.0)	0 (0.0)	0 (0.0)
Local pain	5 (11.6)	3 (6.8)	8 (9.2)
Numbness	21 (48.8)	1 (2.3)	22 (25.3)
Redness	3 (7.0)	0 (0.0)	3 (3.5)
Swelling	11 (25.6)	2 (4.6)	13 (14.9)
Tenderness on palpation	8 (18.6)	1 (2.3)	9 (10.3)
Tingling	2 (4.7)	1 (2.3)	3 (3.5)
Unexpected			
Frostbite	3 (7.0)	0 (0.0)	3 (3.5)

Cryo, cryoneurolysis.

the target tissue and a reduced risk of neuritis, dysesthesia, and neuromas,[20] whereas major blood vessels have also shown resistance to freezing temperatures.[28] The freezing temperatures can affect cutaneous tissue, and three patients observed a small lesion, after treatment, corresponding to a superficial frostbite. The injury did not require treatment, although caution is advised to avoid proximity to cutaneous tissue. Heating pads and hydrodissection were used to minimize its occurrence.

There was a reduction in pain scores 14 days after cryoneurolysis; however, the change was not significantly different compared to the sham group. This contrasts with previous case studies, using different pain models suggesting cryoneurolysis to be effective, and highlights the need for randomized controlled trials to establish the efficacy of treatment. In another randomized controlled trial, Radnovich *et al.*[15] showed a significant reduction in pain scores after cryoneurolysis after 30 to 90 days compared to sham. The results in the current study show a significant difference between groups at the 6-month timepoint, although this was a secondary outcome, and further research is needed. In addition to the intention-to-treat analysis, per-protocol analysis was performed including patients who adhered to the full protocol (cryoneurolysis + GLA:D). The per-protocol analysis showed a significant difference in 24-h numeric rating scale pain between groups both at 14 days and at the 6-month follow-up. The difference in pain between the intention-to-treat and

Table 5. Predicted Average Values and Differences in Functional Capacity for ITT Analysis

	Predicted Average Values [95% CI]		Predicted Difference over Time [95% CI]			
Endpoint	**Cryo**	**Sham**	**Cryo**	**Sham**	**Time # Group [95% CI]**	***P* Value**
Sit to stand						
Baseline	10.7 [9.9 to 11.5]	10.6 [9.8 to 11.5]				
14 days	12.0 [11.1 to 12.8	12.5 [11.6 to 13.4]	1.3 [0.4 to 2.3]	1.8 [0.8 to 2.9]	0.5 [–0.9 to 1.9]	0.446
Post GLA:D	13.6 [12.5 to 14.6]	13.9 [12.7 to 15.0]	2.9 [1.7 to 4.1]	3.2 [2.0 to 4.5]	0.3 [–1.4 to 2.0]	0.697
6 months	14.1 [13.1 to 15.1]	13.4 [12.4 to 14.5]	3.4 [2.3 to 4.5]	2.8 [1.7 to 4.0]	–0.6 [–2.2 to 1.0]	0.447
12 months	14.3 [13.4 to 15.3]	14.0 [12.9 to 15.1]	3.7 [2.6 to 4.7]	3.4 [2.2 to 4.6]	–0.3 [–1.9 to 1.3]	0.723
Walk 40 m						
Baseline	26.6 [25.7 to 27.4]	26.0 [25.2 to 26.9]				
14 days	25.0 [24.1 to 25.8]	26.1 [25.1 to 27.0]	–1.6 [–2.6 to –0.6]	0.0 [–1.0 to 1.1]	1.6 [0.2 to 3.0]	*0.025*
Post GLA:D	23.7 [22.6 to 24.8]	25.0 [23.9 to 26.2]	–2.8 [–4.0 to –1.6]	–1.0 [–2.2 to 0.2]	1.8 [0.1 to 3.5]	*0.040*
6 months	24.3 [23.3 to 25.3]	25.7 [24.7 to 26.8]	–2.3 [–3.4 to –1.2]	–0.3 [–1.4 to 0.9]	2.0 [0.4 to 3.6]	*0.015*
12 months	25.2 [24.3 to 26.2]	25.7 [24.6 to 26.8]	–1.3 [–2.4 to –0.2]	–0.4 [–1.6 to 0.8]	0.9 [–0.7 to 2.5]	0.260
MVC						
Baseline	285.6 [267.4 to 303.7]	284.6 [265.9 to 303.2]				
14 days	285.8 [267.3 to 304.4]	289.1 [269.1 to 309.2]	0.3 [–20.7 to 21.2]	4.6 [–17.9 to 27.0]	4.3 [–26.5 to 35.0]	0.784
Post GLA:D	318.0 [294.0 to 342.0]	335.9 [311.3 to 360.5]	32.4 [6.5 to 58.4]	51.3 [24.7 to 77.9]	18.9 [–18.3 to 56.0]	0.319
6 months	318.5 [296.4 to 340.6]	318.1 [295.5 to 340.8]	32.9 [8.7 to 57.2]	33.6 [8.7 to 58.4]	0.6 [–34.1 to 35.3]	0.972
12 months	301.7 [279.9 to 323.4]	335.2 [312.1 to 358.4]	16.1 [–7.8 to 40.0]	50.7 [25.4 to 75.9]	34.6 [–0.2 to 69.3]	0.051

The values represent the estimated outcomes for each group, accounting for the random effects and covariates included in the mixed effects model, *i.e.*, baseline pain level, time, group. The interaction "Time # Group" signifies the difference between groups across time. 95% CIs are displayed in brackets and statistically significant *P* values ($P < 0.05$) are *italicized.*

Cryo, cryoneurolysis; GLA:D, Godt Liv med Artrose I Danmark (Good Life with osteoArthritis in Denmark); ITT, intention to treat; MVC, maximum voluntary contraction.

per-protocol analyses at 14 days might indicate that the patients in the sham group, who did not complete GLA:D, had higher pain scores 14 days after intervention, although the results are not clear. Other than numeric rating scale pain scores at 24 h, all other results were similar between intention-to-treat and per-protocol analyses. One explanation for why the differences between groups are not clearer in the intention-to-treat analysis might be high variability in pain levels and large placebo effects, especially in this patient group.[29–31] This, combined with a moderate pain level and high quality of life, makes it difficult to differentiate the treatment effect between groups. Finally, at the 14-day timepoint, it is unclear when treatment effects can be expected to occur and for how long.[32] Residual procedural effects may obscure early differences, while gradual neurophysiological changes and delayed modulation of central sensitization in chronic pain may require more time to emerge. This might in part explain the observed effects at 6 months, assuming they represent a true effect rather than a type 1 error.

Various pain indices were evaluated,[33] and least pain experienced improved at 14 days, post GLA:D and at 6 months. Least pain might be a measure of patient ability to function generally, reducing mild, constant/occasional pain,[34] affecting daily activities and quality of life. While we did not observe a change in quality of life, we did observe improvements in both walk speed and anxiety levels at the 6-month follow-up. These changes should be interpreted with caution because they are secondary outcomes and not corrected for multiple comparisons.

Importantly, the current study included an exercise program, suggested to improve pain and function, but the effect remains controversial. Previous studies have suggested that a reduction in pain might improve adherence and force-generating capacity, thus improving the overall effects of exercise. This study evaluated the effect of an exercise program performed within 6 months of treatment, leveraging the period in which pain hypothetically was reduced. We did not observe improved adherence in the cryoneurolysis group compared to sham; however, the reduction in pain was higher and more persistent. The study design does not clarify whether exercise or cryoneurolysis caused these changes, but the combination may be worth further exploration.

The overall improvements were limited. This is an early study, and additional rigorous randomized controlled trials are needed. There is no consensus on the optimal protocol, *i.e.*, freeze duration, number of cycles, thaw duration. To ensure patient safety, the current study used a conservative protocol consisting of a single 2.5-min freeze cycle at each nerve target. There is limited literature on an optimal protocol[20] when using cryoneurolysis for analgesia. Future studies should investigate more aggressive protocols, with multiple cycles and longer freeze durations. Other key factors are target identification, visualization, and probe control. The current study used ultrasound and transcutaneous electrical nerve stimulation to improve safety and precision. Other studies utilize a blind approach with

Table 6. Knee Injury and Osteoarthritis Outcome Score

	Predicted Average Values [95% CI]		Predicted Difference over Time [95% CI]			
Endpoint	**Cryo**	**Sham**	**Cryo**	**Sham**	**Time # Group [95% CI]**	***P* Value**
KOOS QoL						
Baseline	38.7 [34.7 to 42.8]	39.0 [34.8 to 43.1]				
14 days	46.6 [42.5; 50.7]	49.2 [44.7 to 53.6]	7.9 [3.4 to 12.4]	10.2 [5.4 to 15.0]	2.3 [–4.2 to 8.9]	0.491
Post GLA:D	49.0 [44.2 to 53.7]	52.3 [47.1 to 57.5]	10.3 [5.2 to 15.3]	13.3 [7.8 to 18.8]	3.1 [–4.4 to 10.5]	0.424
6 months	53.4 [49.0 to 57.8]	51.7 [46.8 to 56.6]	14.7 [9.9 to 19.4]	12.7 [7.5 to 17.9]	–1.9 [–8.9 to 5.1]	0.596
12 months	51.5 [46.9 to 56.0]	54.5 [49.3 to 59.7]	12.7 [7.8 to 17.6]	15.5 [10.0 to 21.0]	2.8 [–4.6 to 10.1]	0.457
KOOS pain						
Baseline	56.0 [52.3 to 59.7]	56.3 [52.6 to 60.1]				
14 days	67.7 [64.0 to 71.4]	68.6 [64.6 to 72.6]	11.8 [7.5 to 16.0]	12.3 [7.7 to 16.8]	0.5 [–5.7 to 6.7]	0.873
Post GLA:D	72.3 [67.9 76.7]	66.7 [61.9 to 71.5]	16.3 [11.5 to 21.1]	10.4 [5.2 to 15.6]	–5.9 [–13.0 to 1.2]	0.104
6 months	72.0 [68.0 to 76.1]	69.9 [65.5 to 74.4]	16.0 [11.5 to 20.6]	13.6 [8.6 to 18.6]	–2.4 [–9.1 to 4.3]	0.477
12 months	69.6 [65.4 to 73.8]	67.7 [62.9 to 72.3]	13.6 [9.0 to 18.3]	11.3 [6.1 to 16.6]	–2.3 [–9.3 to 4.7]	0.523
KOOS ADL						
Baseline	63.4 [60.0 to 66.8]	63.8 [60.3 to 67.3]				
14 days	73.2 [69.8 to 76.7]	76.7 [73.0 to 80.4]	9.8 [6.0 to 13.7]	12.9 [8.8 to 17.0]	3.1 [–2.6 to 8.7]	0.288
Post GLA:D	77.9 [73.9 to 82.0]	77.7 [73.3 to 82.0]	14.5 [10.1 to 18.9]	13.8 [9.1 to 18.6]	–0.7 [–7.2 to 5.8]	0.830
6 months	78.4 [74.7 to 82.1]	75.4 [71.3 to 79.6]	15.0 [10.9 to 19.1]	11.6 [7.1 to 16.2]	–3.4 [–9.5 to 2.8]	0.279
12 months	76.2 [72.4 to 80.1]	75.8 [71.4 to 80.2]	12.8 [8.6 to 17.1]	12.0 [7.2 to 16.7]	–0.9 [–7.3 to 5.5]	0.792
KOOS symptoms						
Baseline	59.1 [55.5 to 62.7]	59.2 [55.5 to 62.9]				
14 days	68.0 [64.4 to 71.7]	68.7 [64.7 to 72.7	8.9 [4.6 to 13.3]	9.5 [4.8 to 14.2]	0.6 [–5.8 to 7.0]	0.857
Post GLA:D	70.1 [65.8 to 74.4]	67.5 [62.9 to 72.2]	11.0 [6.1 to 15.9]	8.3 [3.0 to 13.6]	–2.7 [–9.9 to 4.5]	0.462
6 months	73.0 [69.0 to 77.0]	71.1 [66.7: 75.6]	13.9 [9.3 to 18.5]	11.9 [6.9 to 17.0]	–2.0 [–8.8 to 4.9]	0.571
12 months	70.4 [66.3 to 74.5]	67.6 [62.9 to 72.4]	11.3 [6.5 to 16.1]	8.4 [3.1 to 13.7]	–2.9 [10.1 to 4.3]	0.426
KOOS sport						
Baseline	28.3 [23.8 to 32.8]	28.4 [23.8 to 33.1]				
14 days	39.4 [34.9 to 44.0]	41.6 [36.7 to 46.5]	11.1 [5.8 to 16.5]	13.2 [7.4 to 18.9]	2.0 [–5.9 to 9.9]	0.617
Post GLA:D	40.5 [35.2 to 45.9]	41.4 [35.5 to 47.4]	12.3 [6.2 to 18.4]	13.0 [6.3 to 19.6]	0.7 [–8.3 to 9.7]	0.877
6 months	44.3 [39.4 to 49.3]	40.9 [35.4 to 46.4]	16.0 [10.3 to 21.8]	12.4 [6.2 to 18.7]	–3.6 [–12.1 to 4.9]	0.404
12 months	45.1 [39.9 to 50.2]	39.0 [33.0 to 44.9]	16.8 [10.9 to 22.7]	10.5 [3.9 to 17.1]	–6.3 [–15.1 to 2.6]	0.165

The table shows the predicted average values and differences from the KOOS questionnaire for the ITT analysis. These values represent the estimated outcomes for each group, accounting for the random effects and covariates included in the mixed effects model, *i.e.*, baseline pain level, time, group. The interaction "Time # Group" signifies the difference between groups across time. "KOOS sport" indicates sport and recreation function. 0 represents extreme knee problems, and 100 represent no knee problems. 95% CIs are displayed in brackets, and statistically significant *P* values ($P < 0.05$) are *italicized*.

ADL, activities of daily living; Cryo, cryoneurolysis; GLA:D, Godt Liv med Artrose I Danmark (Good Life with osteoArthritis in Denmark); KOOS, Knee Injury and Osteoarthritis Outcome Score; QoL, quality of life.

predefined treatment lines[14,15] and other equipment. The optimal strategy is unclear and will also depend on the target nerve. Considering the target nerves in the current study, the anterior femoral cutaneous nerve and the infrapatellar branch of the saphenous nerve might not innervate deeper levels of the knee joint associated with intra-articular processes,[35–37] although studies have shown an analgesic effect. It would be interesting to observe results from studies utilizing other nerve targets such as the study of Panagopoulos *et al.*[38] Increased risks for injury to motor neurons such as the common fibular nerve or neurovascular structures should be considered to ensure balance between treatment efficacy and safety. The anterior femoral cutaneous nerve and infrapatellar branch of the saphenous nerve used in the current study are located away from the above structures while allowing for the use of ultrasound to identify and visualize the target nerve in an outpatient clinical setting. Overall needle placement in relation to the nerve is crucial and possibly a key explanation for the variable results.

Strengths and Limitations

This study is limited by multiple factors. (1) There might be performance and attrition biases, potentially from patients pursuing alternative treatments. (2) There is a risk for multiple testing effects. (3) There is no consensus on the approach to treatment with cryoneurolysis. The cautious approach in this study aimed to reduce risks but may also have limited the effectiveness of the treatment. (4) The GLA:D program was managed by external physiotherapists at specialized GLA:D clinics. The investigators in this study did not take part and thus had limited control and access to data. (5) Finally, the study was performed during a period with COVID-19 and its appertaining shutdowns, which led to patients abstaining from GLA:D, follow-up visits,

Table 7. Additional Patient-reported Outcomes for ITT Analysis

Endpoint	Predicted Average Values [95% CI]		Predicted Difference over Time [95% CI]		Time # Group [95% CI]	*P* Value
	Cryo	Sham	Cryo	Sham		
EQ-5D index						
Baseline	0.84 [0.81 to 0.87]	0.83 [0.80 to 0.86]				
14 days	0.86 [0.83 to 0.88]	0.87 [0.84 to 0.90]	0.02 [−0.02 to 0.06]	0.04 [0.00 to 0.08]	0.02 [−0.03 to 0.07]	0.438
Post GLA:D	0.87 [0.83 to 0.90]	0.87 [0.84 to 0.91]	0.03 [−0.01 to 0.07]	0.04 [−0.00 to 0.08]	0.01 [−0.05 to 0.07]	0.768
6 months	0.88 [0.84 to 0.91]	0.86 [0.82 to 0.89]	0.04 [0.00 to 0.08]	0.02 [−0.02 to 0.07]	−0.02 [−0.07 to 0.04]	0.552
12 months	0.84 [0.80 to 0.87	0.87 [0.84 to 0.91]	0.00 [−0.04 to 0.04]	0.04 [−0.00 to 0.08]	0.04 [−0.02 to 0.10]	0.155
EQ-5D VAS						
Baseline	77.5 [74.7 to 80.2]	77.2 [74.4 to 80.0]				
14 days	80.3 [77.5 to 83.2]	80.8 [77.8 to 83.9]	2.9 [−0.6 to 6.4]	3.6 [−0.1 to 7.3]	0.8 [−4.3 to 5.8]	0.772
Post GLA:D	79.1 [75.5 to 82.6]	80.8 [77.1 to 84.5]	1.6 [−2.5 to 5.7]	3.6 [−0.3 to 7.5]	2.0 [−3.9 to 7.9]	0.509
6 months	82.0 [79.0 to 85.1]	80.8 [77.5 to 84.1]	4.6 [0.9 to 8.3]	3.6 [−0.3 to 7.5]	−1.0 [−6.3 to 4.4]	0.728
12 months	81.4 [78.2 to 84.6]	80.5 [76.9 to 84.1]	3.9 [0.1 to 7.8]	3.3 [−0.9 to 7.5]	−0.6 [−6.3 to 5.0]	0.829
Pain Disability Index						
Baseline	18.5 [15.6 to 21.4]	18.8 [16.0 to 21.7]				
14 days	14.8 [11.9 to 17.8]	12.6 [9.4 to 15.8]	−3.7 [−7.3 to −0.0]	−6.3 [−10.1 to −2.5]	−2.6 [−7.9 to 2.7]	0.331
Post GLA:D	11.8 [8.3 to 15.4]	10.8 [7.2 to 14.4]	−6.7 [−10.8 to −2.6]	−8.1 [−12.2 to −3.9]	−1.4 [−7.2 to 4.5]	0.648
6 months	12.7 [9.5 to 16.0]	10.8 [7.4 to 14.2]	−5.8 [−9.7 to −1.9]	−8.1 [−12.1 to −4.1]	−2.3 [−7.9 to 3.3]	0.423
12 months	14.8 [11.4 to 18.2]	11.3 [7.6 to 15.0]	−3.7 [−7.7 to 0.3]	−7.6 [−11.8 to −3.3]	−3.9 [−9.7 to 2.0]	0.196
Tampa Scale of Kinesiophobia						
Baseline	37.0 [35.3 to 38.7]	36.8 [35.1 to 38.5]				
14 days	34.9 [33.1 to 36.6]	34.1 [32.3 to 36.0]	−2.1 [−4.1 to −0.1]	−2.7 [−4.8 to −0.6]	−0.6 [−3.5 to 2.3]	0.691
Post GLA:D	33.3 [31.2 to 35.4]	32.2 [30.0 to 34.4]	−3.7 [−6.0 to −1.4]	−4.6 [−7.0 to −2.2]	−1.0 [−4.3 to 2.4]	0.580
6 months	33.3 [31.5 to 35.2]	32.6 [30.6 to 34.6]	−3.7 [−5.8 to −1.6]	−4.2 [−6.4 to −2.0]	−0.5 [−3.6 to 2.5]	0.733
12 months	34.4 [32.2 to 36.1]	34.4 [32.1 to 36.6]	−2.8 [−5.0 to −0.7]	−2.5 [−4.9 to −0.1]	0.4 [−2.9 to 3.6]	0.819
Patient Health Questionnaire						
Baseline	3.5 [2.8 to 4.2]	3.1 [2.4 to 3.8]				
14 days	2.5 [1.7 to 3.2]	3.0 [2.2 to 3.8]	−1.0 [−2.0 to −0.0]	−0.1 [−1.2 to 0.9]	0.9 [−0.5 to 2.3]	0.224
Post GLA:D	3.1 [2.2 to 4.0]	3.3 [2.3 to 4.3]	−0.4 [−1.5 to 0.7]	0.2 [−1.0 to 1.4]	0.6 [−1.1 to 2.3]	0.503
6 months	2.8 [2.0 to 3.6]	3.1 [2.2 to 3.9]	−0.7 [−1.7 to 0.3]	0.0 [−1.1 to 1.1]	0.7 [−0.8 to 2.2]	0.389
12 months	3 [2.2 to 3.8]	3.2 [2.2 to 4.2]	−0.5 [−1.5 to 0.6]	0.1 [−1.1 to 1.3]	0.6 [−1.0 to 2.2]	0.486
Pain Catastrophizing Scale						
Baseline	13.4 [11.5 to 15.3]	13.4 [11.5 to 15.3]				
14 days	8.9 [7.0 to 10.7]	8.9 [6.9 to 11.0]	−4.6 [−6.7 to −2.4]	−4.5 [−6.9 to −2.2]	0.0 [−3.2 to 3.2]	0.987
Post GLA:D	9.0 [6.7 to 11.3]	8.4 [5.9 to 10.8]	−4.4 [−6.9 to −1.8]	−5.1 [−7.8 to −2.4]	−0.7 [−4.4 to 3.0]	0.701
6 months	8.1 [6.1 to 10.2]	8.9 [6.6 to 11.1]	−5.3 [−7.6 to −2.9]	−4.6 [−7.7 to −2.1]	0.7 [−2.7 to 4.1]	0.690
12 months	10.0 [7.9 to 12.2]	8.2 [5.9 to 10.6]	−3.4 [−5.8 to −1.0]	−5.2 [−7.8 to −2.5]	−1.8 [−5.4 to 1.8]	0.322
Generalized Anxiety Disorder						
Baseline	1.6 [1.1 to 2.1]	1.5 [0.9 to 2.0]				
14 days	0.6 [0.1 to 1.1]	1.2 [0.6 to 1.8]	−1.0 [−1.7 to −0.3]	−0.3 [−1.1 to 0.6]	0.7 [−0.4 to 1.8]	0.185
Post GLA:D	1.8 [1.2 to 2.5]	1.3 [0.6 to 2.0]	0.3 [−0.6 to 1.1]	−0.2 [−1.1 to 0.7]	−0.4 [−1.6 to 0.8]	0.502
6 months	0.9 [0.3 to 1.5]	2.0 [1.3 to 2.6]	−0.7 [−1.5 to 0.1]	0.5 [−0.3 to 1.3]	1.2 [0.1 to 2.3]	*0.039*
12 months	1.7 [1.2 to 2.4]	1.6 [0.9 to 2.3]	0.2 [−0.6 to 1.0]	0.1 [−0.8 to 1.0]	0.0 [−1.2 to 1.2]	0.943

The table shows the predicted average values and differences from patient-reported outcomes for the ITT analysis. These values represent the estimated outcomes for each group, accounting for the random effects and covariates included in the mixed effects model, *i.e.*, baseline pain level, time, group. The interaction "Time # Group" signifies the difference between groups across time. 95% CIs are displayed in brackets and statistically significant *P* values ($P < 0.05$) are *italicized*.

Cryo, cryoneurolysis; GLA:D, Godt Liv med Artrose I Danmark (Good Life with osteoArthritis in Denmark); ITT, intention to treat; VAS

and social contact generally. Some uncertainty and data loss were inevitable.

This study's strength is its rigorous design as a blinded, sham-controlled randomized controlled trial with long-term follow-up. The sham treatment closely imitates the cryoneurolysis procedure in terms of physical marks and the sensory experience. A combination of anatomical landmarks, electrical nerve stimulation, and ultrasound imaging was used to accurately locate the nerves, which improves the accuracy and safety of the procedure, and the same surgeon, proficient in identifying neural structures and performing cryoneurolysis, carried out all procedures in this study.

Conclusions

The results of this double-blinded, sham-controlled trial suggest that cryoneurolysis with the tested protocol does not provide a reduction in pain compared to sham at the primary 14-day endpoint in patients with chronic knee osteoarthritis. Further, statistically significant improvements after

6 months were small, were not consistently observed across other pain measures or quality-of-life indices, and were not sustained at 12 months. While the treatment was generally well tolerated, with only minor and transient adverse effects, the data do not establish a clinically relevant, robust benefit.

The study does not support cryoneurolysis as a superior option over sham for managing chronic knee osteoarthritis pain, at least with the tested protocol and patient population. Future research should explore other treatment parameters with multiple or longer freeze cycles, consider alternative nerve targets, and incorporate larger samples and extended follow-ups to clarify the role for cryoneurolysis as a potential adjunct in knee osteoarthritis pain management.

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Competing Interests

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Reproducible Science

The data presented in this study are available on request from the corresponding author (niels-peter.brochner.nygaard@rsyd.dk). The data are not publicly available due to privacy and ethical reasons.

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Supplemental Digital Content

Raw data, https://links.lww.com/ALN/D845

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